Master Product Specification and Implementation Manual: Project UDAAN

An AI-Enabled Vocational Career Guidance and Family Decision-Support Platform

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1. Product Vision, Philosophy, and System Mandate

Project UDAAN is a multi-stakeholder, public-interest educational technology platform designed to transform vocational career guidance in India from a static, one-time recommendation into an interactive, evidence-grounded decision engine.

1.1 Core Problem Statement

India possesses a youth population exceeding 250 million, yet its demographic potential is hindered by structural mismatches between educational streams, vocational training infrastructure, and industrial workforce needs. While platforms such as the Skill India Digital Hub (SIDH) have digitized registration for over 20 million citizens, student dropouts and low enrolment in technical trades persist due to socio-cultural stigma, systemic information asymmetry, and intergenerational household resistance.

In Indian households, career selection is an intergenerational decision. Parents act as primary decision-makers or veto holders. Operating under risk aversion and limited exposure to modern technical trades, parents frequently push youth toward conventional academic degrees, resulting in widespread underemployment alongside acute technical skill shortages.

1.2 System Mission & Value Proposition

UDAAN resolves this intergenerational gap by establishing an interactive decision-support ecosystem. The system enables students to explore, experience, and evaluate vocational careers through physics- and logic-based digital simulations while generating empirical evidence reports for parents. By replacing societal assumptions with verified data on earnings, National Skills Qualifications Framework (NSQF) accreditation levels, and vertical career progression, UDAAN aligns student aspirations with family consent and market demands.

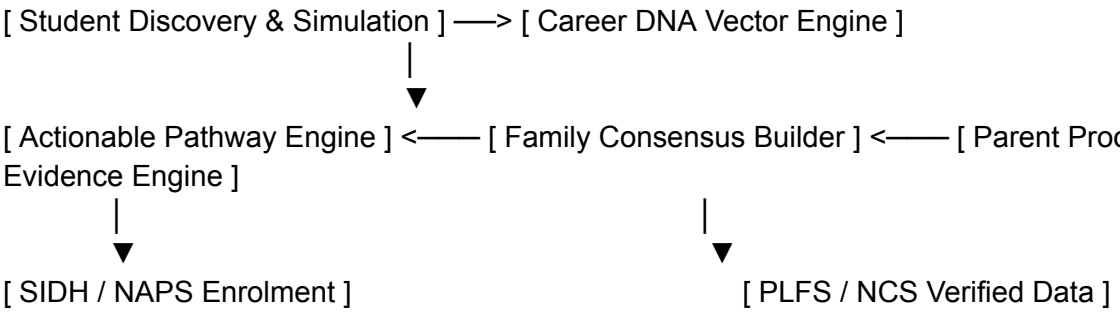
2. Target Audience and Stakeholder Persona Matrix

The platform serves six distinct user personas across five dedicated, role-based portal views, ensuring tailored UI layouts while operating on a single source of truth.

User Segment	Primary Operational Objectives	Core Socio-Economic Friction Points	Digital Literacy & Access Profile	Key Platform Interventions
School Students *(Classes 8–12)*	Discover latent aptitudes; explore practical trades; select secondary vocational electives.	Peer pressure toward general degrees; opacity regarding technical trade realities.	High mobile familiarity; low technical vocabulary; shared family devices.	Interactive Holland RIASEC Career DNA; gamified trade micro-simulations.
Vocational Candidates *(ITIs/Polytechnics)*	Identify specialized trades; track NAPS apprenticeships; plan self-employment.	Perceived ceiling on career growth; social stigma regarding manual labor.	Moderate-to-high smartphone usage; localized vernacular preferences.	Actionable 8-stage roadmaps; stackable NSQF qualification maps.
Parents & Guardians	Safeguard social standing; ensure child's financial security; evaluate training costs.	Fear of unaccredited centers; doubt over trade wage stability and career progression.	Low-to-moderate digital literacy; reliance on voice, video, and simple text.	Parent Proof Mode; 16-point empirical dossiers; What-If financial comparators.

Institutional Counsellors	Manage student cases; provide human intervention for decision paralysis.	High student-to-counsellor ratios; lack of objective, standardized student diagnostic data.	High desktop/tablet proficiency; structured case management requirements.	Consented student profile summaries; XAI decision trees; priority referral queues.
Teachers & Coordinators	Track class career interests; guide students toward fitting vocational streams.	Outdated industrial demand knowledge; administrative overhead.	Moderate desktop/laptop literacy; structured workflow preferences.	Institutional analytics dashboards; group guidance toolkits; engagement tracking.
Government Administrators	Monitor regional skill demand; identify systemic barriers; optimize training infrastructure.	Data fragmentation across ministries; delayed policy feedback loops.	High administrative data needs; spatial GIS mapping requirements.	Spatial GIS gap maps; reported barrier taxonomies; differential privacy aggregate indicators.

3. Core Functional Modules and Feature Architecture



3.1 Dynamic Career DNA Profiling Engine

The Career DNA engine constructs an evolving representation of student competencies, interests, and real-world constraints, rejecting static diagnostic classifications.

The Career DNA profile vector $P_u(t)$ for user u at time t is modeled as:

$$P_u(t) = \left[R_u(t), C_u(t), L_u(t), S_u(t), E_u(t)\right]$$

Where:

- $R_u(t) \in R^6$ represents the normalized Holland RIASEC vector (Realistic, Investigative, Artistic, Social, Enterprising, Conventional), where $\sum_{i=1}^{6} R_{u,i} = 1$.
- $C_u(t) \in [0,1]^k$ denotes practical competencies evaluated during micro-simulations.
- $L_u(t)$ captures financial, temporal, and spatial constraints (e.g., maximum travel distance, stipend requirement).
- $S_u(t)$ records self-reported skill confidence across numerical, spatial, and digital domains.
- $E_u(t)$ tracks exploration history, including completed simulation telemetry and repeated views.

3.2 Reverse Career Search Architecture

Instead of mandating an initial job choice, the Reverse Career Search matches candidates based on lifestyle requirements. The Compatibility Index $S(u,j)$ for user u and trade role j is calculated as:

$$S(u,j) = w_1 \cdot f_{interest}(R_u, T_j) + w_2 \cdot f_{skill}(C_u, K_j) - \sum_m \lambda_m \cdot g_m(L_u, D_{j,m})$$

Where T_j and K_j define the required interest and skill profiles for trade j, g_m measures constraint violations (such as travel distance or training budget limits), and w_1, w_2, λ_m are calibrated weighting parameters.

3.3 Career Exploration Laboratory (Interactive Micro-Simulations)

The laboratory provides hands-on micro-simulations replicating technical tasks within a digital environment. Diagnostic telemetry updates the candidate's Career DNA profile in real time.

Trade Simulation	Task Objective & Scenario	Telemetry Mechanics	Performance Indicators	NSQF Alignment
Solar PV Technician	Diagnose energy drop in array; adjust tilt angle for optimal irradiance.	Drag-and-drop circuit tracing; tilt angle vs. irradiance slider adjustments.	Angular optimization accuracy; safety isolation compliance time.	Solar PV Installer (Suryamitra), NSQF Level 4.
Domestic Electrician	Isolate distribution board faults; wire a two-way switch circuit.	Virtual component placement; wire terminal mapping; lock-out tag-out sequence.	Circuit continuity verification; safety lock-out execution.	Electrician Domestic, NSQF Level 4 / ITI CTS Trade.
Healthcare Assistant	Conduct preliminary intake triage; communicate vitals protocol.	Branching dialogue selection; digital sphygmomanometer reading sequence.	Protocol compliance accuracy; patient empathy index.	General Duty Assistant, NSQF Level 3/4.
Automotive Technician	Diagnose four-stroke IC engine non-starting issue; check fuel lines vs. starter.	Interactive engine schematic exploration; pressure testing execution.	Diagnostic logic efficiency; correct tool selection.	Auto Service Technician, NSQF Level 4.

Customer Operations	Resolve retail customer return escalation under time constraints.	Multi-choice branching dialogue; POS system query navigation.	De-escalation effectiveness; inventory system speed.	Retail Customer Service Associate, NSQF Level 3.
Digital Data Services	Format and clean rural agricultural inventory spreadsheet data.	Formula entry; duplicate filtering; automated chart generation wizard.	Transformation accuracy; syntax execution correctness.	Domestic Data Entry Operator, NSQF Level 4.

3.4 Parent Proof Mode & 16-Point Evidence Engine

When a candidate selects a pathway, the system generates a shareable Parent Evidence Report covering 16 verification parameters grounded in official data sources:

1. **Occupational Scope:** Plain-language explanation of daily responsibilities.
2. **Core Competencies:** Practical skills acquired during instruction.
3. **Educational Prerequisites:** Formal entry requirements (Class 8, 10, or 12 pass).
4. **Training Duration:** Total hours and calendar months required.
5. **NSQF Qualification Rating:** Official NCVET qualification level (Levels 1–8).
6. **Verified Course Fee Structure:** Subsidized government routes vs. private costs.
7. **Financial Assistance:** PMKVY 4.0 funding and NAPS stipend eligibility details.
8. **Institutional Accreditation:** List of local Government ITIs and NCVET-affiliated centers.
9. **Geographic Hiring Intensity:** Regional market demand indicators.
10. **Empirical Salary Benchmarks:** Verified entry, mid-career, and senior wage distributions drawn from Periodic Labour Force Survey (PLFS) data.
11. **Self-Employment Viability:** Capital requirements and MUDRA micro-finance linkages.
12. **Vertical Career Progression:** Career growth maps detailing transitions to master technician, supervisor, or B.Voc degree options.
13. **Workplace Safety Regulations:** Mandatory industrial safety protections.
14. **Lateral Pathway Options:** Transition opportunities into adjacent technical sectors.
15. **Parent FAQ Matrix:** Addressing social status and financial security concerns.
16. **Verifiable Data Sources:** Direct verification links to Skill India Digital and NCS portals.

3.5 What-If Scenario Simulator

Families can evaluate side-by-side comparative models over a 5-year horizon:

- **Option A (Conventional):** 2 years higher secondary general stream + 3 years general BA degree.
- **Option B (Direct Vocational):** 2 years ITI Craftsman Training Scheme (CTS) + 1 year NAPS apprenticeship.
- **Option C (Short-Term Skilling):** 6 months PMKVY 4.0 NSQF Level 4 training + immediate entry-level placement + part-time diploma.

The engine models direct costs, opportunity costs, apprenticeship stipends, and cumulative projected earnings, establishing financial clarity.

3.6 Family Consensus Builder

The consensus builder aligns student choices and parent concerns:

1. **Independent Assessment:** Student logs goals; parent logs financial and safety concerns.
2. **Cross-Mapping Engine:** Identifies shared priorities and maps points of divergence to evidence reports.
3. **Action Plan Generation:** Output categorizes items into Fully Agreed, Needs Verification, Simulation Assignment, or Counsellor Escalation. Student private logs remain protected.

3.7 Actionable Pathway Engine

The pathway engine guides execution through an 8-stage sequence:

- **Stage 1: Discovery:** Explore domain roles and Career DNA alignment.
- **Stage 2: Experiential Testing:** Complete trade micro-simulations.
- **Stage 3: Prerequisites:** Verify educational eligibility and accreditation.
- **Stage 4: Financial Planning:** Identify PMKVY 4.0 free training or stipends.
- **Stage 5: Official Enrolment:** Direct candidate registration via Skill India Digital Hub API.
- **Stage 6: Assessment:** Undergo NCVET / Sector Skill Council evaluation.
- **Stage 7: Apprenticeship Routing:** Transition into paid on-the-job training via NAPS.
- **Stage 8: Career Progression:** Access advanced technical certifications or B.Voc bridge courses.

4. UI/UX Design System and Multi-Role Information Architecture

4.1 Design System Token Specifications

The visual design system enforces accessibility, clear visual hierarchy, and high contrast for low-cost mobile displays.

Design Token	Hex Value	Purpose & Application	Contrast Ratio (WCAG 2.1 AA)
Primary Navy	#17324D	Top headers, primary navigation containers, institutional badges.	12.6:1 over white surface
Primary Teal	#087F8C	Core interactive controls, active state highlights, CTA buttons.	4.8:1 over background
Bright Teal Accent	#19B5A5	Progress bars, completed pathway nodes, success highlights.	3.1:1 over deep navy
Warm Saffron Accent	#F2A541	High-value milestone badges, focus highlights, key callouts.	3.0:1 over deep navy
Neutral Background	#F6F8FB	Main application surface background.	N/A (Base Surface)
Surface Base	#FFFFFF	Component card surfaces, modal containers.	N/A (Base Surface)
Primary Text	#182230	Main body typography, section titles, primary metrics.	14.1:1 over white surface

Secondary Text	#657386	Supporting labels, contextual help text, secondary info.	4.6:1 over white surface
Status Success	#21865B	Assessment completion, verified course badges.	4.7:1 over white surface
Status Warning	#B7791F	Pending parent reviews, verification pending status.	4.5:1 over white surface
Status Error	#C43D4B	Validation errors, high-risk flags, simulation warnings.	5.2:1 over white surface

Typography utilizes **Inter** for Latin script interface elements paired with **Noto Sans Devanagari** for Hindi text, structured on an 8-pixel grid system.

4.2 Multi-Role Portal Navigation Architecture

- **Student Portal View:** Dashboard, Career DNA Summary, Career Exploration Lab, Actionable Pathways, Family Space, My Progress.
- **Parent/Guardian Portal View:** Family Overview, Shared Career Evidence Reports, What-If Comparator, Family Consensus Builder, Financial Planning Resources.
- **Counsellor Portal View:** Priority Referral Queue, Consented Case Logs, XAI Decision Trees, Session Tracking & Notes.
- **Teacher/Institution Portal View:** Aggregate Class Interest Metrics, Group Guidance Toolkits, Institutional Referral Tracking.
- **Government Administrator Portal View:** Spatial GIS Demand Maps, Barrier Analytics Taxonomy, SIDH Enrolment Trackers, Policy Outcome Reports.

4.3 Onboarding and Authentication Flow

1. **Welcome Surface:** Low-bandwidth entry screen showcasing platform value pillars (*Discover*, *Experience*, *Plan Together*).
2. **Role Selection Panel:** Role selection cards outlining permissions and privacy controls.
3. **Language Selector:** Multi-script switcher supporting Scheduled Indian languages, launching with complete Hindi and English interfaces.
4. **Low-Friction Login:** Mobile number verification via OTP or institutional SSO (Aadhaar login is non-mandatory).

5. **Demographic Ingestion:** District location, education level, and preferred study mode collection.
6. **Interactive Platform Tour:** 4-step guide detailing Career DNA profiling and simulation tools.
7. **Unbundled Consent Panel:** Granular opt-in toggles governing AI guidance, counsellor data sharing, parent linkage, and aggregate research analytics.

5. Conversational AI, Explainability, and Risk Protocols

5.1 Retrieval-Augmented Generation (RAG) Architecture

The conversational AI interface acts as a supportive career companion. To prevent hallucinated qualifications or wage data, queries are processed via a Retrieval-Augmented Generation (RAG) pipeline. The system retrieves verified facts from vector databases containing indexed NCVET, PMKVY 4.0, and Labour Ministry documentation prior to model inference.

5.2 Explainable AI (XAI) Engine

Recommendations are fully interpretable. The platform integrates Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) paired with Natural Language Generation:

- **Student Perspective:** Plain-language natural language summaries explaining recommendation logic without technical jargon.
- **Counsellor Perspective:** LIME feature attributions and decision tree weights explaining recommendation factors during case reviews.
- **Administrator Perspective:** Global SHAP feature attribution heatmaps detailing systemic interest drivers across regional demographic cohorts.

5.3 Behavioral Analysis and 4-Level Risk Escalation

The AI monitors interaction patterns (such as repeated decision hesitation or financial distress signals) strictly as dialogue cues to trigger supportive UI interventions, avoiding unwarranted psychological profiling.

Risk Classification	**Operational Criteria & System Dialogue Signals**	**Automated System Response**	**Human Escalation Protocol**

Level 0: Normal Exploration	Standard inquiries regarding course fees, job duties, NSQF levels.	AI provides grounded answers with career cards and simulation links.	Self-guided navigation; no human escalation triggered.
Level 1: Decision Paralysis	Repeated goal changes; decision hesitation loops; confusion between trades.	AI slows interaction pace, generates trade-off summaries, and surfaces decision tools.	Banner suggests booking a session with an institutional counsellor.
Level 2: Substantial Distress	Student expresses severe anxiety over family pressure or impending academic dropout.	AI provides supportive messaging, pauses recommendations, and surfaces Human Support Modal.	Requests student consent to generate a Priority Referral Ticket for an assigned counsellor.
Level 3: Immediate Safety Concern	Explicit communication of self-harm intent or severe abuse.	AI immediately presents verified national crisis helplines (Tele-MANAS, KIRAN) and halts career chat.	Automated alert sent to designated institutional emergency desk per safeguarding rules.

6. Relational Data Schema and Security Specification

6.1 Database Architecture & Schema Matrices

Persistence is managed via PostgreSQL with the pgvector extension, maintaining relational integrity alongside vector search capabilities.

Entity Table	Primary Key	Foreign Keys	Key Attributes & Data Properties	Access Control & Privacy Boundary
users	user_id (UUID)	district_id	mobile_hash, role_enum, preferred_lang, created_at	System Service Access Only; Mobile numbers hashed at ingestion.
student_profiles	profile_id (UUID)	user_id -> users	education_level, preferred_study_mode, budget_limit	Student Read/Write; Linked Parent Read-Only (Consented fields).
career_dna	dna_id (UUID)	profile_id -> student_profiles	riasec_vector (Vector6), competency_scores (JSONB)	Student Read/Write; Counsellor Read-Only (On Referral).
simulations	sim_id (UUID)	nsqf_level_id	trade_domain, scenario_config (JSONB), telemetry_schema	Public Read-Only.

simulation_attempts	attempt_id (UUID)	profile_id, sim_id	task_completion_time, safety_score, telemetry_logs (JSONB)	Student Read-Only; Aggregate Analytics Engine.
guardian_links	link_id (UUID)	student_id, parent_id -> users	consent_status, shared_permissions (JSONB), linked_at	Student & Linked Parent Read-Only; Requires explicit opt-in.
referrals	referral_id (UUID)	student_id, counsellor_id	risk_level_enum, summary_text, status_enum, created_at	Assigned Counsellor Read/Write; Student Read-Only.
barriers_analytics	barrier_id (UUID)	district_id	barrier_category_enum, reported_count, anonymized_cohort	Public Admin Read-Only; Enforces $N \geq 50$ differential privacy.

6.2 Data Privacy and Regulatory Compliance

- **DPDP Act Alignment:** Mandates explicit parent/guardian consent flows for minor students under age 18.
- **Data Minimization:** Restricts storage to necessary operational parameters and excludes mandatory Aadhaar linkage.
- **Self-Service Privacy Controls:** Enables students to inspect, export, or permanently delete their profile data.

- **No Commercial Profiling:** Prohibits student data monetization, ad targeting, or transfer to third-party data brokers.

7. Public Administration Analytics and Spatial GIS Engine

7.1 Spatial GIS Infrastructure Gap Engine

The administrative dashboard monitors vocational demand and infrastructure capacity. The spatial mapping engine calculates an Infrastructure Gap Index (IGI_d) for district d:

$$IGI_d = \frac{D_{student,d}}{S_{accredited,d} + \epsilon}$$

Where $D_{student,d}$ represents student interest metrics for specific trades within district d, and $S_{accredited,d}$ measures the capacity of active, accredited ITIs and PMKVY skill hubs. High gap scores flag regions requiring targeted skilling infrastructure investment.

7.2 Barrier Analytics Taxonomy

Anonymized user survey data and parent concern logs populate an empirical Barrier Taxonomy:

- **Socio-Cultural / Status Stigma:** Parental preference for general degrees despite poor market outcomes.
- **Spatial Access Barriers:** Lack of accredited skill centers within commuting distance.
- **Financial Misconceptions:** Unawareness of PMKVY 4.0 free skilling and NAPS stipends.
- **Qualification Opacity:** Confusion regarding NCVET certificate validity and degree bridge pathways.

Differential privacy constraints mandate a minimum threshold of $N \geq 50$ user interactions per cohort before displaying aggregate administrative analytics.

8. Implementation Roadmap and Verification Metrics

8.1 Phased Delivery Roadmap

- **Phase 1 (Hackathon Prototype):** Core onboarding, Career DNA profiling, 6 interactive micro-simulations, Parent Proof Mode, and baseline RAG conversational guidance.
- **Phase 2 (State Pilot & SIDH Integration):** Live API integration with Skill India Digital Hub, NAPS apprenticeship routing, SHAP/LIME explanation pipelines, and counsellor portal deployment.
- **Phase 3 (Nationwide Rollout):** Full expansion across 22 Scheduled Indian languages, voice navigation support, offline PWA synchronization, and statewide GIS spatial gap analytics.

8.2 Comprehensive Metric Evaluation Matrix

Stakeholder Persona	Primary Outcome Metric	Evaluation Formula / Standard	Target Success Threshold
Student Engagement	**Simulation Completion Rate**	$\frac{CompletedMicro-Simulations}{InitiatedMicro-Simulations} \times 100$	$\ge 75\%$ across active users.
Student Progression	**Pathway Exploration Depth**	$\frac{UsersReachingStage4+Pathways}{TotalRegisteredStudents} \times 100$	$\ge 60\%$ of onboarding candidates.
Parent Engagement	**Evidence Report Utilization**	$\frac{ParentsViewingProofReports}{TotalLinkedParentAccounts} \times 100$	$\ge 80\%$ of linked parent profiles.
Family Alignment	**Consensus Plan Execution**	$\frac{FamiliesCompletingNext-StepPlan}{FamiliesEnteringConsensusModule} \times 100$	$\ge 50\%$ agreement rate.
Counsellor SLA	**Referral Resolution Time**	$Timestamp_{FirstContact} - Timestamp$	< 24 hours for Level 2 referrals.
Public Governance	**District Gap Coverage**	Spatial density of aggregate barrier reports per state	100% district mapping coverage.

9. Master Execution Summary

Project UDAAN bridges student aspirations, family consent, and industrial workforce needs. By replacing societal stigma with verified evidence and hands-on micro-simulations, UDAAN establishes a scalable blueprint for public-interest educational technology in India.